

#27 Solar Output Forecaster

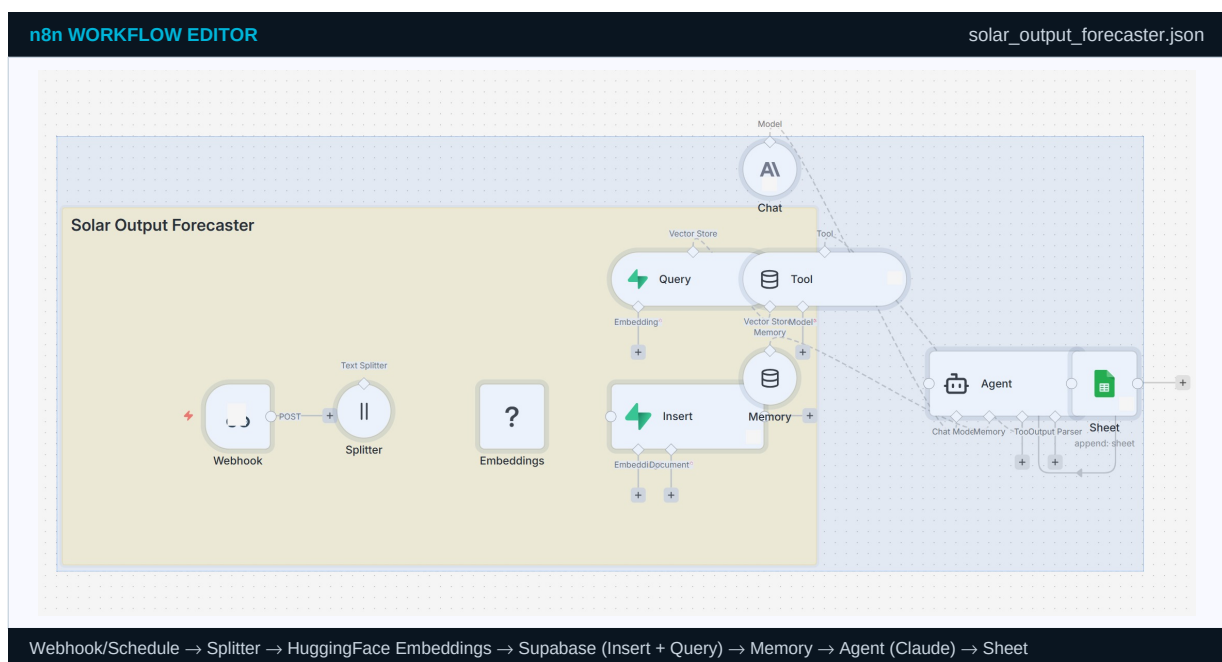
Energy /
Renewable
Forecasting

Predict tomorrow's solar kWh output — optimise battery storage and grid export scheduling

BUSINESS PROBLEM

Solar operators over-commit grid export without accurate next-day forecasts, incurring balancing charges. This workflow fetches weather API data, matches it against similar historical generation days using vector search, and delivers 24-hour hourly forecasts with confidence bounds for automated grid scheduling.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Schedule POST — daily 22:00 weather forecast for next-day grid planning

```
{
  "site_id": "solar-site-04",
  "forecast_date": "2024-06-16",
  "weather_forecast": [
    {
      "hour": "06:00",
      "irradiance_wm2": 180,
      "cloud_pct": 15,
      "temp_c": 28
    },
    {
      "hour": "12:00",
      "irradiance_wm2": 980,
      "cloud_pct": 5,
      "temp_c": 41
    },
    {
      "hour": "18:00",
      "irradiance_wm2": 320,
      "cloud_pct": 35,
      "temp_c": 38
    }
  ],
  "panel_count": 480
}
```

OUTPUT — Hourly forecast array — written to Sheets for EMS integration

```
{
  "json": {
    "site_id": "solar-site-04",
    "daily_total_kwh": 3847,
    "grid_export_kwh": 2200,
    "battery_kwh": 1647,
    "mape_7d": "4.2%",
    "hourly": [
      {
        "hour": "06:00",
        "kwh": 87,
        "confidence": "LOW"
      },
      {
        "hour": "12:00",
        "kwh": 445,
        "confidence": "HIGH"
      },
      {
        "hour": "18:00",
        "kwh": 298,
        "confidence": "MEDIUM"
      }
    ]
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Schedule at 22:00 — next-day forecast run	Daily trigger generates forecast in time for grid scheduling decisions at midnight.
2	⌘ Splitter — weather and performance doc chunks	Historical generation reports and weather correlation analysis chunked for RAG retrieval.
3	■ HuggingFace Embeddings — weather pattern vectors	Open-source model encodes weather profiles for similar historical day lookup.
4	■ Supabase Insert — store weather forecast	24-hour forecast stored: {hour, irradiance_wm2, cloud_cover_pct, temp_c, wind_mps} per hour.
5	■ Supabase Query — similar historical days	Retrieves 10 most similar past weather days with their actual generation outputs.
6	■ Vector Tool — historical generation data	Agent fetches actual kWh generated on matched historical days for weighted forecast.
7	■ Window Memory — 7-day accuracy tracking	Tracks forecast vs actual. Flags MAPE above 8% — may indicate panel soiling.
8	■ Claude — forecast calculation	Similarity-weighted historical average + weather regression → hourly kWh with confidence intervals.
9	■ AI Agent — structured forecast output	[[{hour, forecast_kwh, lower_ci, upper_ci, confidence, key_driver}]] for all 24 hours.
10	■ Google Sheets — EMS integration	Forecast written to Sheets. Energy management system reads via API for battery scheduling.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
HuggingFace API Token	huggingface.co/settings/tokens → New token read scope	HuggingFace API
Supabase URL + Anon Key	app.supabase.com → Settings → API	Supabase API
Anthropic (Claude) API Key	console.anthropic.com → API Keys	Anthropic API
OpenWeatherMap API Key	openweathermap.org/api → One Call API 3.0	HTTP Header Auth

SAMPLE RUN — what this workflow processed and produced

Scenario	A 480-panel rooftop solar installation feeds a commercial building. The EMS needs next-day generation forecasts by midnight to optimise battery charge scheduling.
Input received	22:00 trigger on 15 June. OpenWeatherMap returned 24-hour forecast: mostly clear, peak irradiance 980 W/m2 at 12:00, some afternoon cloud from 15:00.
What the workflow found	Supabase found 10 historically similar days. Average generation: 3,820 kWh. Afternoon cloud risk adjusted 15:00 forecast down with wider confidence intervals. 7-day MAPE: 4.2%.

Output produced	24-hour forecast in Sheets by 22:04. EMS scheduled battery charge to 80% by 10:00. Actual output next day: 3,901 kWh vs forecast 3,847 kWh — 1.4% error. No grid balancing charges.	
JSON: solar_output_forecaster.json	GitHub: trm-solar-output-forecaster	openweathermap.org/api